

Dynamic Model Control & Optimized System Analysis of a New Designed Unmanned Aerial Vehicle - TSF

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its empirical nature due to the convergent efforts of the mathematicians to solve the nonlinear problems involved in the calculation methods in aerodynamics and also thanks to the invention of the very fast machines of calculation and the development of data processing.

In this present work, a graphical simulation of a force around an UAV and description of the flight dynamics and its control system is presented. The first phase of this work consists in using plans of construction of this UAV to create a solid model which specifies its external geometry in Solid Works. This solid model is then exported into a programme like Autodesk Inventor of the grid generation in order to return its geometry adapted to aerodynamic calculations and tested in various flying conditions. A computer code of computational fluid dynamics (CFD) is used to obtain estimates of the coefficients of lift and drag to 10 meters per second. A study of the independence of the grid is then carried out to determine the precision of the grid of calculation used for analysis CFD and other airflow stress simulations. Secondly the flight dynamics is put into account where necessary data is extracted for the control system to act according to the attitude of the flight.

II. ADVANCEMENTS

In normal UAVs like the Reaper and Panther it contains the universal flight mode but lacks full safe modes of landing at high terrain places during emergency thus weakening their part. Now combining both the features brings to an ultimate UAV which proves a lot during the war field. But it's not that simple to design such a model as it need a lot of mechanical factors which falls good during aerodynamic conditions as a small force point at the non-pressure points will lead to damage and crash landing. There have been jet fighters like the Jet Harrier with VTOL and Flight modes but they are for attack. And there have been a lot of designs for Surveillance with both these features but no one has yet brought down to a feasible and marketable application. We have worked on the mechanical design now which stands to be feasible as it being tested on the aerodynamic use as the mechanical design has been tested well. The aerodynamic and mechanical simulation are described here which will prove to its feasibility. More from the mechanical side, it can stop very

I. INTRODUCTION

The unmanned aerial vehicle (UAV) is a small plane which flies in an autonomous way with only one virtual pilot on board. That does not mean that the plane is completely out of control once it leaves the track. The presence of a human being in the loop of control is essential. But the operator is on a station on the ground, either on ground, or at sea or in a control center. He can intervene at any time, examine the situation and the rules of engagement, and stop the mission if necessary. The concept of an UAV resonates in several research laboratories of any developed country. The first instigators of this type of the projects were the soldiers. Indeed, the armies of many countries prefer to rather send UAV's for the recognition in the high-risk zones than man. This type of flying machine can take much various missions, extending from the control of the scientific experiments to the intelligence collecting the monitoring during the day or the night. The numerical simulation in the field of aerodynamics is relatively recent research tools. The specialists raise the question of knowing if the field of aerodynamics is purely theoretical or numerical, they estimate however that it is theoretical, because of the many numerical tests necessary for the stage to insufficiencies of the knowledge of the calculation methods. Since a score of years, one can say that numerical aerodynamics lost much of

Keywords- UAV, Drones, Flight mode, VTOL Mode, Booster Motors, Control system, Tricopter, Kinematics.

Abstract— In this paper we will discuss about control system & system analysis of a newly designed model of UAV which has been specially made for Vertical Take-off & Landing (VTOL) and as well as a mode which converts it to a high speed flight mode UAV which has been never been designed yet with a better feasibility factor. The old forms of Tricopters have only the VTOL advantage but fail to escape from sight when the enemy falls into the view, thus lacking the stealth mode. And apart from mechanical side, it has features like GPS tracking, Live video streaming, High point stability, Stealth modes and more. Features of the UAV, it had fail safe modes like, if the drone is with one damaged motor, then the back or front part actuators can individually bring the drone back to base station acting as the backup and main drive.



Fig.2 Vertical Lift off and Landing Mode

This UAV as shown in Fig. 3 is 2 meters long and provided with an elevator wing of the scale of .40 m. the profile of this wing is of type NACA 2412 with a cord of 0.236 m, stalled on the fuselage of 3° with a dihedral of 2°. The elevator group of the shape of T reversed is profiled with Naca 0012 and attached to the wings by two beams which give also the name of an UAV to tristar. These UAV's aspire of a remarkable stability to the no desirable movements of roll. This UAV is designed to have a booster motors which will be embarked behind the fuselage.

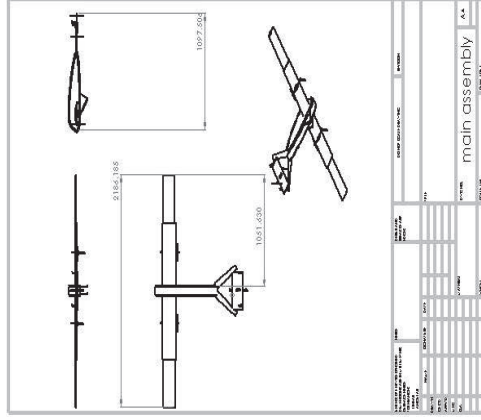


Fig. 3 2D CAD Model

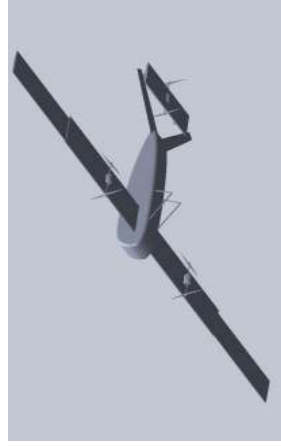


Fig.1 Flight Mode

Now during the flight mode the main wings rotate 90 degrees to horizontal position and the secondary wings also retract or extend which brings it to the flight mode. The booster motors also rotate to horizontal positions and applying the necessary speed to the glider.

III. DESCRIPTION OF THE UAV

As you can see in Fig.1, it is in the normal extended flight mode now. In the design, there are two movable wings at the end of which are connected the brushless motor and the tri blade. These wings are connected to the high torque servo motors which are installed in the hull of the drone. These wings stay vertical during the take-off or landing mode. Adjacent the point of brushless motor is there is another servo motor whose shaft is connected to the secondary wing which unfolds during the flight mode. These secondary wings help for gliding in the air thus saving the power of the battery. Now to the back of the drone, we have the booster brushless motor which gives the speed to the UAV. They are also movable to a resolution of 90 degrees. During VTOL mode Fig.2, it is also in the vertical position, and this feature of having it vertical brings to the fact that it won't restrict the direction of take-off. This will cut down the wind and pause and instability coming in its way.

swiftly during flight modes in the case of emergency. It does not have to take a gliding turn when needed, it just needs to stop at the point and turn at the point itself thus saving the flight time. In brief, in this abstract we'll discuss about some of the mechanical design, and its simulation and analysis. The photos are provided for a well virtual creation and analysis.

IV. SYSTEM DESIGN SIMULATION

A. Airfoil Wind Tunnel Test

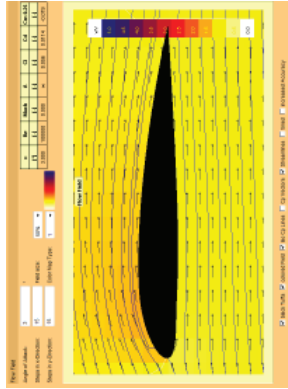


Fig. 4 Velocity Ratio Analysis

Several compilations were carried out for various angles of incidence to determine the lift and drag. Each case was carried out at the same speed: 10 m/s. The angle of attack extended from 3 to 18 degrees. It is a parabolic type tendency similar to those found on the planes in general. In particular, one notes the increase in the drag with the increase in the angle of incidence. This curve is presented with drag of the isolated wing which equips the UAV in question and which was obtained by wind tunnel test. The two diagrams follow the same tendency separately which is normal because variation of the drag between the two curves corresponds to the drag of the remainder of the UAV which varies very slightly with the incidence. The current NACA airfoil proves to be having a low turbulence and a better lift factor with the angle of attack set at 3degrees. The velocity ratio is balanced properly. And the pressure generation around the airfoil is distributed symmetrically converging into a stable airfoil design.

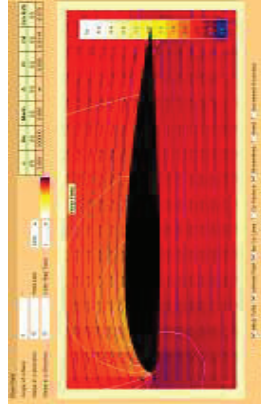


Fig.5 Pressure Co-efficient Analysis

B. Eppler & NACA Analysis

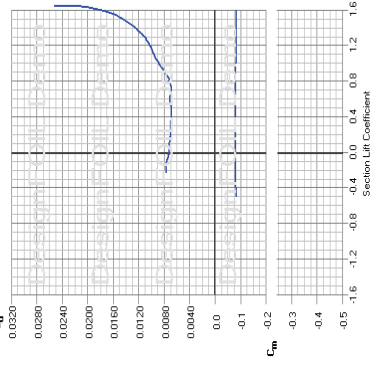


Fig.6 NACA Analysis of Cd vs Cl

An agreement of the two curves is noted in the linear zone which is explained by the fact that it is the wing which contributes in a great proportion to the total lift produced on the plane. The other parts of the plane, in particular the stabilizer contribute slightly towards weak and moderate incidence. However the lift of the isolated wing exceeds that of the plane for the bigger angles of incidence. This is due to the fact that the Reynolds number of the tests of the isolated wing is much higher than that of the plane tested numerically which leads to higher maximum C_m for the isolated wing. Numerical calculations show well the phenomenon of unhooking of the UAV which is located at approximately 3 degrees. It should be noted that this incidence which appears is weak only seemingly because it should be mentioned that the reference of measurement of the angle of attack is the base of the fuselage and that the wing is fixed with 4 degrees, flow return is noted on almost the totality of the wing, in particular, it is much attenuated in the central part.

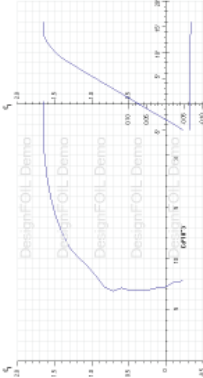


Fig.7 Eppler Analysis of Cd vs Cl

This effect is expected because the wing is not twisted and the great deflections of the air are perceptible in this area.

The separation of the flow recorded on the nose of the fuselage still persists with this angle of incidence, flow recorded on the level of the wing root with the fuselage with 15° of incidence.

C. CFD Analysis of Fuselage & Airfoil

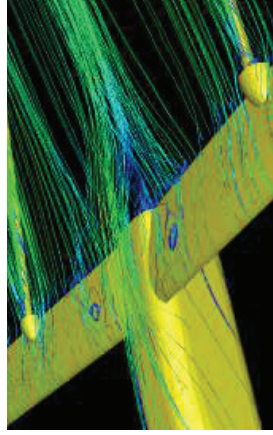


Fig.8 Air flow around Fuselage & Airfoil

It is a representation of the streamlines of the flow inside the internal zone of the boundary layer around the UAV which is completely pressure building. It is a very useful diagram for the draftsman of the aerodynamic forms because it reveals the design defects characterized by the appearance of the zones of flow separated color blue as shown in the Fig.9 on the wing and on the level from the bases from fastener from the wings. Fig. 9 shows the two marginal swirls which release from salmons of the two wings of the UAV to the passage of the flow for the incidence 15°, the size of the marginal swirl for such an incidence is considered to be considerable. Moreover, one notes the complexity of the flow around the plane for an incidence judged like the beginning of turbulence.

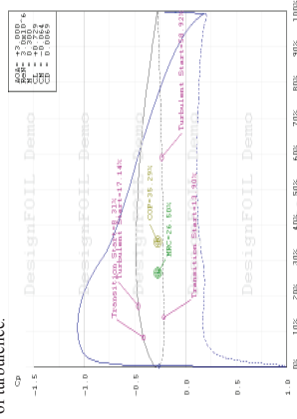


Fig.9 Turbulent Analysis

The trailing edge leads to a better flow with a slight generation of turbulence at the top camber line which is obvious as there generation of pressure built at that position.

D. Wing Stress & Loading Analysis

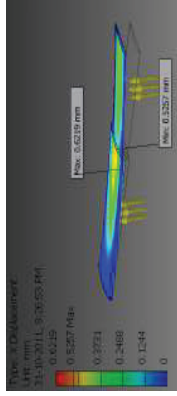


Fig.10 Wing Loading Effect

While both aerodynamic and structural metrics are included in the wing design, the analysis are not explicitly coupled in an aeroelastic sensor, stable deflections of the wing due to air loads are assumed to be small, since large ones will lead to buckling failure. Aerodynamic performance of the wing is studied using the fixed wing stress analysis. Aerodynamic analysis computes the pressure distribution over the wing, as well as the concomitant lift, lift-to-drag ratio (L/D) and the static stability margin. An arc length method based nonlinear deviation tool is used to evaluate the structural performance of the wing and to determine maximum flying velocity that the wing can withstand without buckling or failing (laminar failure with factor of safety = 1.5) under aggressive flight loads.

Experimental validation of the numerical procedure used for determining maximum flying velocity has been carried out and can be found elsewhere. An analytical approach based on classical laminate plate theory and strain superposition (initial chord-wise strains due to root airfoil flattening and additional span-wise strains due to incremental rolling) is used to study the stresses developed in the composite wing during the rolling process for storage with a safety factor of 1.5 is used to check first ply failure of the composite wing to determine minimum safe storage diameter.

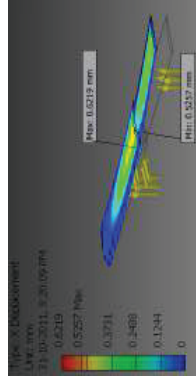


Fig.11 Thrust & Loading Effect

Taking advantage of problem symmetry, we model only one wing half with the appropriate symmetric boundary conditions at the root airfoil. And appropriate results have been found as the stress analysis proved correct and the wing was able to sustain the pressure at the trailing edge as we

rarely see reddened part on the airfoil. Reaching to a corrected figured value achievable to 10m/s or 20m/s.

V. GENERAL ARCHITECTURE

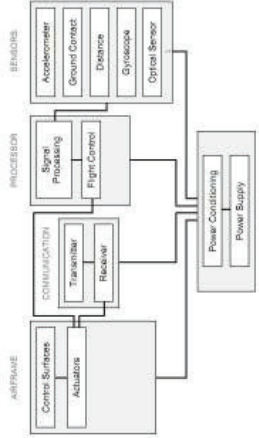


Fig.12 Architecture

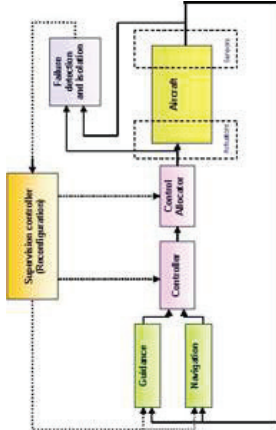


Fig.13 Control Structure

and velocity, whereas the dynamic model defines the relation governing the applied forces and the resulting accelerations

B. Reference frames

Before getting into the equations of kinematics and dynamics of the tricopter, it is necessary to specify the adopted coordinate systems and frames of reference, as well as how transformations between the different coordinate systems are carried out. The use of different coordinate frames is essential for identifying the location and attitude of the tricopter in six degrees of freedom (6 DOF). For example, in order to evaluate the equations of motion, a coordinate frame attached to the tricopter is required. However, the forces and moments acting on the tricopter, along with the inertial measurement unit (IMU) sensor values, are evaluated with reference to the body frame. Finally, the position and speed of the tricopter are evaluated using GPS measurements with respect to an inertial frame located at the base station.

Thus, three main frames of reference are adopted, as shown in Figure 14:

1. The inertial frame, $\mathcal{F}_i = (\vec{x}_i, \vec{y}_i, \vec{z}_i)$, is an earth-fixed coordinate system with the origin located on the ground, for example, at the base station. By convention, the x-axis points towards the north, the y-axis points towards the east, and the z-axis points towards the center of the earth.
2. The body frame $\mathcal{F}_b = (\vec{x}_b, \vec{y}_b, \vec{z}_b)$, with its origin located at the center of gravity (COG) of the tricopter, and its axes aligned with the tricopter structure such that the x-axis $\vec{x}_b$ is along the arm with front motor, the y-axis $\vec{y}_b$ is along the arm with right motor, and the z-axis $\vec{z}_b = \vec{x}_b \times \vec{y}_b$, where 'x' denotes the cross product.

3. The vehicle frame, $\mathcal{F}_v = (\vec{x}_v, \vec{y}_v, \vec{z}_v)$, is the inertial frame with the origin located at the COG of the tricopter. The vehicle frame has two variations, $\mathcal{F}_\phi$ and $\mathcal{F}_\theta$. $\mathcal{F}_\phi$ is the vehicle frame, $\mathcal{F}_v$, rotated about its z-axis $\vec{z}_v$ by an angle ϕ so that $\vec{x}_v$ and $\vec{y}_v$ are aligned with $\vec{x}_b$ and $\vec{y}_b$, respectively. $\mathcal{F}_\theta$ is frame $\mathcal{F}_\phi$ rotated about its y-axis, $\vec{y}_v$, by a pitching angle, θ , such that $\vec{x}_v$ and $\vec{z}_v$ are aligned with $\vec{x}_b$ and $\vec{z}_b$, respectively.

Translation and rotation matrices are used to transform one coordinate reference frame into another desired frame of reference. For example, the transformation from $\mathcal{F}_i$ to $\mathcal{F}_v$ provides the displacement vector from the origin of the inertial frame to the center of gravity (COG) of the tricopter. Also, the transformation from $\mathcal{F}_v$ to $\mathcal{F}_b$ is rotational in nature, therefore yielding the roll, pitch and yaw angles.

$$R_{\mathcal{F}_\phi}^{\mathcal{F}_b}(\phi) = R_{\mathcal{F}_b}^{\mathcal{F}_\phi}(\theta) = R_{\mathcal{F}_\phi}^{\mathcal{F}_b}(\psi) = I$$

Therefore,

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{\mathcal{F}_b} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & s\phi c\theta \\ 0 & -s\phi & c\phi c\theta \end{bmatrix}_{\mathcal{F}_b} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{\mathcal{F}_v}$$

It follows that,

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{\mathcal{F}_b} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & s\phi c\theta \\ 0 & -s\phi & c\phi c\theta \end{bmatrix}_{\mathcal{F}_b} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{\mathcal{F}_v}$$

VII. DYNAMIC MODEL OF THE VEHICLE

The vehicle is equipped with 2 wings wings that are mounted on the front and an elevator at back of the vehicle and can be rotated from vertical to horizontal positions. Figure 15 below shows the aerial vehicle in vertical flight modes.

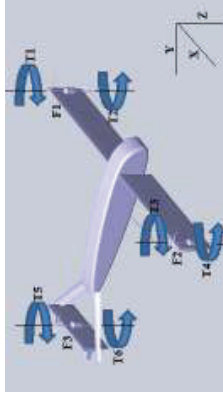


Fig.15 Aerial Vehicle in Vertical Flight Modes

With this wing configuration, the vehicle's airframe transforms into a tricopter like structure if the wings are at the vertical position and into an airplane like structure if the wings are at the horizontal position. To keep the control complexity on a minimum level, the rotations of the wings are used as attitude control inputs in addition to motor thrust inputs of the vehicle in horizontal flight mode. Therefore the need of additional control surfaces that are put on trailing edges of the wings on a regular airplane are eliminated. Two wings on the front can be rotated independently to act like the ailerons while wing at the back are rotated together to act like the elevator. This way the control surfaces of a regular plane in horizontal flight mode are mimicked with minimum number of actuators.

A. Vertical Flight Mode

In vertical flight mode, the mechanical structure of the vehicle is very similar to a tricopter vehicle and the dynamics of the vehicle can be approximated by the model given below: Fig. 15 Aerial Vehicle in Vertical Flight Configuration. The two reference frames given in Fig. 15 are body fixed reference frame $B : (Ox_b, y_b, z_b)$ and earth fixed inertial reference frame $W : (Ox_w, y_w, z_w)$. Using this model, the equations describing the position and attitude of the vehicle are obtained by relating the 6 DOF kinematic equations with the dynamic equations. The position and linear velocity of the vehicle's center of mass in world frame are described as,

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}, \quad \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix}$$

The attitude and angular velocity of the vehicle in world frame are given as,

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}, \quad \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

where, ϕ, θ, ψ are named roll, pitch and yaw angles respectively. The equations for the transformation of the angular and linear velocities between world and body frames are given in equations (1) and (2):

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = T_{bw}(\phi, \theta, \psi) \cdot \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

where

$$T_{bw}(\phi, \theta, \psi) = R_x(\phi)R_y(\theta)R_z(\psi)$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = E(\phi, \theta) \cdot \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Where

$$E(\phi, \theta) \cdot \omega = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\theta) & \sin(\theta) \cos(\theta) \\ 0 & -\sin(\theta) & \cos(\theta) \cos(\theta) \end{bmatrix}$$

The dynamic equations for the forces and torques acting on the vehicle's center of gravity are given in equations (3) and (4):

$$\mathbf{F}_{tot}^v = M_b \dot{\mathbf{V}}_b + \omega_b \times (M_b \cdot \mathbf{V}_b)$$

Where

$$M_b = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix}$$

$$M_{tot}^v = I_n \omega_b + \omega_b \times (I_n \cdot \omega_b)$$

Where

$$I_n = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}$$

I_x, I_y and I_z are moment of inertias in the body reference frame. The total force F_v tot acting on the vehicle's center of gravity is the sum of the lifting forces F_v cg created by the rotors, the gravity F_g and the aerodynamic forces F_a which is considered as a disturbance, namely

$$\mathbf{F}_{tot}^v = \mathbf{F}_{cg}^v + \mathbf{F}_g + \mathbf{F}_a^v$$

Where

$$\mathbf{F}_{cg}^v = \begin{bmatrix} F_x^v \\ F_y^v \\ F_z^v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -(F_1 + F_2 + F_3) \end{bmatrix}$$

$$\mathbf{F}_g = \begin{bmatrix} -\sin(\theta) \\ \sin(\phi) \cos(\theta) \\ \cos(\phi) \cos(\theta) \end{bmatrix} \cdot mg$$

The total torque M_v tot acting on the vehicle's center of gravity (cog) is the sum of the torques M_v cg created by the rotors and the aerodynamic torques M_a which is again considered as a disturbance, namely

$$M_{tot}^v = M_{cg}^v + F_a^v$$

Where

$$M_{cg}^v = \begin{bmatrix} M_x^v \\ M_y^v \\ M_z^v \end{bmatrix} = \begin{bmatrix} I_x & -I_{xy} & I_{xz} \\ I_{xy} & I_y & -I_z \\ I_{xz} & I_z & \lambda_3 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

I_x and I_z are the distances of rotors to the center of gravity of the vehicle in x and y directions. Note also that $I_i = I_i r_i$ for $i = 1, 2, 3$ and sum of torques created by the rotors result in a yaw moment along the z axis in vertical flight mode:

$$M_z^v = \sum_i T_i$$

B. Horizontal Flight Mode

In horizontal flight mode, the mechanical structure of the aerial vehicle forms the airframe of a four winged airplane with two wings on the front and two wings at the back. The aerodynamic lift and drag forces that are considered as external disturbance in vertical flight mode need to be taken into account in dynamic modeling for horizontal flight. Fig. 16 shows the aerial vehicle in horizontal flight mode. The lift forces F_{il} and drag forces F_{id} for $i = 1, 2, 3$ generated by each wing are obtained from the equations:

$$F_{li}^i = -\frac{1}{2} c_{L_i}(\theta_i) \rho A \cdot V^2$$

$$F_{di}^i = -\frac{1}{2} c_{D_i}(\theta_i) \rho A \cdot V^2$$

where the angle of attack for each wing is given by q_i . Note that the lift coefficient $c_{L_i}(q_i)$ and the drag coefficient $c_{D_i}(q_i)$ are not constant variables, but functions of angle of attack q_i for each wing. The total force F_h tot acting on the vehicle's center of gravity is the sum of the forces F_h cg created by the rotors, the gravity F_g , the lift and drag forces generated by the wings.

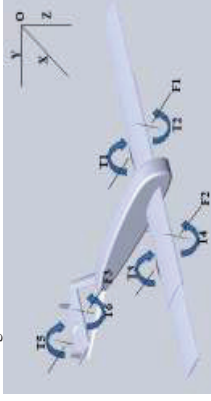


Fig. 16 Aerial Vehicle in Horizontal Flight Configuration

F_w and the aerodynamic forces F_h a which is considered as a disturbance, namely

$$\mathbf{F}_{tot}^h = \mathbf{F}_{cg}^h + \mathbf{F}_g + \mathbf{F}_w + \mathbf{F}_a^h$$

where

$$\mathbf{F}_{cg}^h = \begin{bmatrix} F_x^h \\ F_y^h \\ F_z^h \end{bmatrix} = \begin{bmatrix} F_1 + F_2 + F_3 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_a = \sum_i F_{di}^i + \sum_i F_{li}^i = \begin{bmatrix} -(c_{D_i}(\theta_i) + c_{D_i}(\theta_2) + 2c_{D_i}(\theta_3)) \frac{\rho A}{2} V^2 \\ 0 \\ -(c_{L_i}(\theta_i) + c_{L_i}(\theta_2) + 2c_{L_i}(\theta_3)) \frac{\rho A}{2} V^2 \end{bmatrix}$$

The total torque M^h tot acting on the vehicle's center of gravity is the sum of the torques M_h cg created by the rotors, M_w created by the drag/lift forces of the wings and the aerodynamic torques M_a which is again considered as a disturbance, namely

$$M_{tot}^h = M_{cg}^h + M_w + M_a^h$$

Where

$$M_{cg}^h = \begin{bmatrix} M_x^h \\ M_y^h \\ M_z^h \end{bmatrix} = \begin{bmatrix} -\lambda_1 & -\lambda_2 & -\lambda_3 \\ 0 & 0 & 0 \\ I_x & -I_y & I_z \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

$$M_w = \begin{bmatrix} -(c_{L_i}(\theta_1) - c_{L_i}(\theta_2)) \frac{\rho A}{2} V^2 I_{x_i} \\ -(c_{L_i}(\theta_1) + c_{L_i}(\theta_2) - 2c_{L_i}(\theta_3)) \frac{\rho A}{2} V^2 I_{l_i} \\ -(-c_{D_i}(\theta_1) + c_{D_i}(\theta_2)) \frac{\rho A}{2} V^2 I_{y_i} \end{bmatrix}$$

Note that the sum of torques created by the rotors result in a roll moment along the x axis in horizontal flight mode

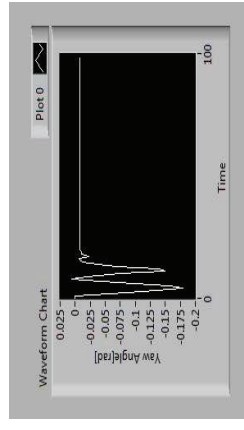
$$M_x^h = \sum_i T_i$$

The dynamic equations obtained for 6 DOF rigid body transformation of the aerial vehicle in inertial reference frame W are also valid for the dynamic model in horizontal flight mode:

$$\mathbf{F}_{tot}^h = M_b \dot{\mathbf{V}}_b + \omega_b \times (M_b \cdot \mathbf{V}_b)$$

$$M_{tot}^h = I_n \dot{\omega}_b + \omega_b \times (I_n \cdot \omega_b)$$

VIII. SIMULATION RESULTS



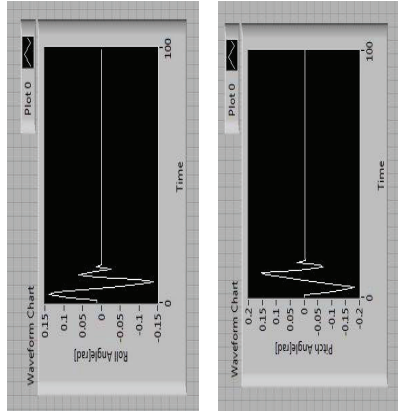


Fig. 17(a)(b)(c) Attitude control of the vehicle under no disturbance condition

Fig. 17 show the variation of the position and attitude state variables for the reference inputs $P_r = (3.3, -15)^\circ$ and $y_r = 0$ under ideal operating conditions without any external disturbance. Note that position and angle references are tracked with negligible steady state errors.

IX. CONTROL LOOPS

For TSF, a PID controller was chosen due to its proven capability in similar systems and its ease of implementation. Because the vehicle was a new design, it was immediately apparent that the time to develop, build, test, and refine the design of this vehicle would be substantial. Thus, an effective yet relatively simple control scheme was ideal, because it minimized the time required to solve this important and complex problem. A number of similar applications and vehicles which used a PID controller were found and it was discovered that the performance of this type of controller would be adequate to complete the competition objectives. Additionally, this type of controller allowed retuning based on observable behavior. Once the gains were set near the optimal values it was possible to tune the vehicle empirically, which reduced the reliance on the (potentially incomplete) dynamic model and provides a measure of robustness.

In any PID controller, the system input is comprised of three terms which are related to the difference (error) between the current system state and a desired state. The proportional (P) term multiplies the error by a gain value and uses this as an input. This increases the influence of this term when the current state is further away from the desired state. The

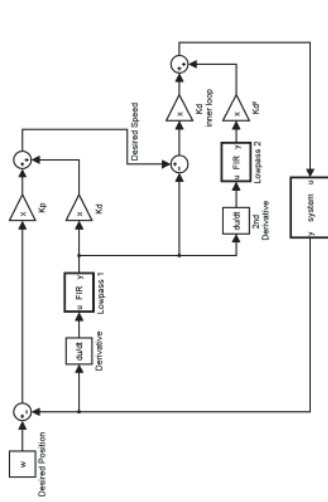


Fig.18 Filtered Control structure

integral (I) term uses the integral of the error over time multiplied by a gain value to influence the input. This generally has a gradual effect that reduces steady state error. The derivative (D) term, takes the derivative of the error and multiplies this by a gain value to influence the input based on how the error of the state is changing. This has a dampening effect which decreases oscillations in the system response. TSF use four PID controllers in each of its roll, pitch, yaw, and height maneuvers.

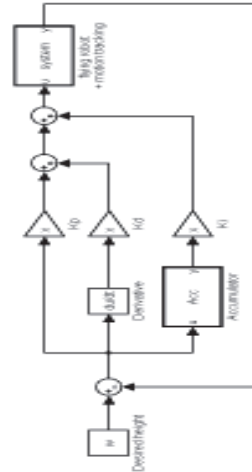


Fig.19 Conditioning Loop

The onboard controllers are three independent PD loops, one for each rotational axes (roll, pitch and yaw). Angular velocities measured by the gyroscopes and computed relative velocities are used as inputs. The angles are derived by integrating the sum of the output of one gyroscope and an external control input for the respective axis. Without an external input signal the calculated integral represents the angle the flying robot has turned in the respective axis. Looking at the closed loop and disregarding measurement noise and integration errors, this means that the robot will always keep its current orientation. The integrated angles can be shifted by an external control input. As a result, the robot's orientation changes

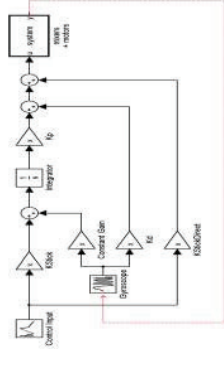


Fig.20 PID loop

proportionally to the input. Its movements are controlled by steering it to a certain orientation and keeping this orientation for a certain time. Due to measurement noise and discrete integration the integrated angles drift about ± 3 degrees per minute. However, this drift can be easily compensated by a human pilot or an autonomous external position control such as a motion capture system. The controller implementations have been optimized for shortest possible execution time and robustness in almost every flight situation. The loop is interrupt triggered, which enables stable time constants for integration and filtering. By using the AVR's internal ADCs at a high sampling rate, fixpoint arithmetics only, runtime optimized FIR filter implementations and interrupt driven communication to update the motor speeds, we achieved a system running at a control frequency of 8MHz.

X. GROUND STATION CONTROL

TSF vehicle status can be monitored through the graphical user interface (GUI). The GUI reports critical flight data such as vehicle altitude, battery life, and communication signal strength as well as the mission state. With this information, we can predict what issues could possibly arise allowing us to better react to a situation and reduce risk as much as possible.



Figure 21 Ground Station Control GUI.

XI. CONCLUSION

The present study was undertaken to observe the aerodynamic performances of an UAV by using a CFD

computer code and to find out the system mathematical model. This concept of design of flying machines is judged like an alternative to the wind tunnel tests for a possible reduction of the design cost. This work brought moreover, certain important aerodynamic corrections around this UAV such as the aerodynamic stability of the nose of the fuselage and the setting of the bases of fastener of the beams on the level of the under-surface of the wings with the proper stress analysis done and higher stability of the UAV under different situations.

Several change and simplification were made in the modeling process, including the use of electric brushless motors instead of reciprocating, the constant mass, counter-rotating propeller to eliminate rotating-mass effects, the removal of aileron control, etc. These changes were made in a way that the flight characteristics would not change much. The simulations were conducted several times performing a take-off, hover, cruise and landing maneuver. Softwares like Solidworks, Autodesk Inventor, Design Foil, Matlab, Simulink, Labview and many others have contributed a lot for the analysis and simulations done in this paper. It's been a successful path along.

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REFERENCES

- [1] Boeing, V-22 Osprey, (2008, September 13). Available: <http://www.boeing.com/roscraf/military/v22/index.htm>
- [2] The Bell Eagle Eye UAS, (2008, September 13). Available: <http://www.bellhelicopter.com/en/aircraft/military/bellageye.htm>
- [3] J.J. Dickeson, D. Miles, O. Cidalo, Wells, V.L. Rodriguez, A.A. "Robust LPV H Gain-Scheduled Hover-to-Cruise Conversion for a Tilt-Wing Rotorcraft in the Presence of CG Variations," *American Control Conference, ACC 07*, vol., no., pp.5266-5271, 9-13 July 2007
- [4] F. Kendoul, I. Fantoni, R. Lozano, "Modeling and control of a small autonomous aircraft having two tilting rotors," *Proceedings of the 44th IEEE Conference on Decision and Control, and the European Control Conference*, December 12-15, Seville, Spain, 2005
- [5] Iju et al. U.S. Patent "Bendable wing for micro air vehicle," Patent no.: US 7,331,546 B2 Date of Patent: Feb. 19, 2008.
- [6] Sobieszczanski-Sobieski, J., Haftka, R., "Multidisciplinary Aerospace Design Optimization: Survey of Recent Developments," *Structural and Multidisciplinary Optimization*, Vol. 14, No. 1, pp. 1-23, 1997.
- [7] Johnson, B., Claxton, B., Sanford, B., Jagdale, V., Ijju, P., "Development of a Composite Bendable-Wing Micro Air Vehicle," *AIAA Aerospace Sciences Meeting and Exhibit*, Reno, NV, January 8-11, 2007.
- [8] Sobieszczanski-Sobieski, J., Haftka, R., "Multidisciplinary Aerospace Design Optimization: Survey of Recent Developments," *Structural and Multidisciplinary Optimization*, Vol. 14, No. 1, pp. 1-23, 1997.
- [9] Snyder, D., "The Quad Tiltrotor: Its Beginning and Evolution," *Proceedings of the 56th Annual Forum, American Helicopter Society*, Virginia Beach, Virginia, May 2000.